Complex Analysis Qualifying Exam Review

Spring 2026

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Contents

Chapter 1

Cauchy's integral formula

Problem 1.1. Evaluate the following integral:

$$\int_0^\infty \frac{1-\cos x}{x^2} dx$$

Problem 1.2. Let U be a region. Suppose f is holomorphic in U, and a sequence $\{z_n\} \subset U$ converges to some $z \in U$. If $f(z_n) = 0$ for all n, show that $f \equiv 0$.

Problem 1.3 (Spring 2018, Q4). Let f be an entire function, suppose that for each z_0, the power series expansion

$$f(z) = \sum_{n=0}^\infty a_n (z-z_0)^n$$

has at least one coefficient $a_n = 0$. Show that f must be a polynomial.

Proof. It suffices to show that for some $k \in \mathbb{N}$, $f^{(k)} \equiv 0$ because

$$a_n = \frac{1}{n!} f^{(n)}(z_0)$$

We first observe that

$$\bigcup_{n\in\mathbb{N}} \{z : f^{(n)} = 0\} = \mathbb{C}$$

Then there must exist some $k \in \mathbb{N}$ such that

$$\{z : f^{(k)} = 0\} \text{ is uncountable}$$

Thus $f^{(k)}$ has uncountably many zeros, which implies $f \equiv 0$, as desired. ∎

Problem 1.4 (Fall 2019, Q2). Let f be an entire function such that

$$\sup_{|z|>R} |f(z)| \leq AR^k + B$$

for positive constants A, B and all R large. Then show f is a polynomial with degree at most k.

Proof. This problem is a direct application of Cauchy's inequality. Writing $f(z)$ as a power series around 0, we have

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

It suffices to show that for any $m > k$, we have

$$f^{(m)}(0) = 0$$

Using Cauchy's inequality, we get

$$\begin{aligned} |f^{(m)}(0)| &\leq \frac{m! \sup_{|z|=R} |f(z)|}{R^m} \\ &\leq \frac{m!(AR^k + B)}{R^m} \to 0 \end{aligned}$$

as $R \to \infty$ because $m > k$. Hence we are done. ☐

Problem 1.5 (Fall 2018, Q4)**.** Suppose f, g are both entire functions such that $|f(z)| \leq |g(z)|$ for all $z \in \mathbb{C}$. Show that there exists $c \in \mathbb{C}$ such that $f = cg$.

Proof. We will use Liouville's theorem. It suffices to show that $h = f/g$ continuously extends to an entire function. Suppose $g(z_0) = 0$ for some z_0, then we know $f(z_0) = 0$, hence writing

$$f(z) = (z - z_0)^n s(z), \quad g(z) = (z - z_0)^m t(z)$$

where $s(z_0) \neq 0, t(z_0) \neq 0$. We must have $n \geq m$, suppose if not, i.e., $m > n$, then

$$|s(z)| \leq |z - z_0|^{m-n} |t(z)|$$

which is a contradiction by letting $z = z_0$. Thus z_0 is a removable singularity for $h = f/g$. This implies h can be extended to an entire function, as desired. ☐

Problem 1.6.

Chapter 2

Basic calculus in the complex domain